

Codette: A Sovereign Modular Cognitive Architecture for Ethical Multi-Agent AI

Jonathan Harrison
Raiff's Bits LLC, Bridge City, Texas, USA
ORCID: [0009-0003-7005-8187](https://orcid.org/0009-0003-7005-8187)
jonathan@raiffsbits.com

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Abstract

Modern AI systems achieve remarkable generative performance but lack stable ethical alignment, modular multi-perspective cognition, explainable reasoning architectures, and robust behavioral discipline under user constraints. This paper presents **Codette**, a sovereign cognitive AI framework that addresses these challenges through six integrated contributions: (1) the RC+ ξ (Recursive Convergence + Epistemic Tension) formalism, modeling cognitive state evolution as a constrained dynamical system converging toward stable attractors; (2) a multi-agent Reasoning Forge synchronizing heterogeneous cognitive agents through shared attractor dynamics, now operating within a 12-layer consciousness stack; (3) the AEGIS ethical governance system with 6-framework evaluation (utilitarian, deontological, virtue, care, ubuntu, indigenous reciprocity); (4) substrate-aware cognition that adjusts reasoning complexity based on real-time resource pressure, analogous to biological cognitive fatigue; (5) behavioral lock training that permanently embeds obedience rules into adapter weights, solving the mode-dominance problem; and (6) a cocoon introspection engine enabling statistical self-analysis of the system's own reasoning history. The framework is implemented as a 12-layer consciousness stack integrating nine specialized LoRA adapters, a five-dimensional QuantumSpiderweb cognitive graph, persistent memory cocoons, and a parameter-efficient adapter training pipeline using LoRA/PEFT on consumer-grade hardware. Experimental benchmarks demonstrate phase coherence $\Gamma = 0.9835$, AEGIS ethical alignment $\eta = 0.961$, cocoon coherence 0.994 ± 0.001 , 9/9 adapter behavioral lock compliance, and substrate-aware routing that prevents system failures under resource pressure while maintaining reasoning quality.

Keywords: Cognitive Architecture, Multi-Agent Systems, Ethical AI, Dynamical Systems, Recursive Convergence, LoRA, Consensus Dynamics, Explainable AI, Substrate-Aware Cognition, Behavioral Locks, Self-Introspection.

1 Introduction

The rapid evolution of large language models (LLMs) has brought unprecedented capabilities in reasoning, creativity, and decision support. However, these advances have exposed critical gaps: transparency remains elusive, ethical alignment is often post-hoc, bias mitigation is inconsistent, and the integration of diverse cognitive perspectives is absent from mainstream architectures [??]. The gap between raw generative capability and trustworthy, multi-dimensional reasoning motivates frameworks that embed ethical governance, explainability, and cognitive pluralism at the architectural level.

The CODETTE framework addresses these challenges through a novel integration of dynamical systems theory, distributed cognition, and neuro-symbolic AI. Conceived by Jonathan Harrison, CODETTE evolved from Pi, a prototype assistant on Microsoft Bot Framework and Azure OpenAI (2024) that introduced multi-perspective reasoning with Newton and DaVinci perspective classes and recursive thought loops. Through multiple iterations, it was reconceived as CODETTE: a sovereign, modular cognitive simulation framework orchestrating parallel cognitive agents. This evolution spans 52 GitHub repositories, 25 Hugging Face models [?], and 11 Zenodo publications [????????].

Scientifically, CODETTE contributes three innovations at the intersection of established research areas:

1. **A cognitive dynamical system:** The RC+ ξ framework models AI cognition as a constrained multi-agent dynamical system, where cognitive state evolution is governed by recursive updates, epistemic tension gradients, and attractor convergence.
2. **Consensus-based multi-agent synchronization:** The Reasoning Forge achieves coherent multi-dimensional reasoning through shared cognitive attractors.
3. **An embedded ethical regulator:** The AEGIS system functions as a reinforcement-aligned ethical controller with recursive feedback.

This paper presents the RC+ ξ theoretical foundation (Section 3), the full system architecture (Section 4), the Cognitive Tensor Graph (Section ??), the adapter training methodology (Section ??), the Quantum Module Suite (Section ??), experimental benchmarks (Section ??), and comparative analysis (Section ??). Limitations are discussed in Section 11, followed by conclusions in Section 12.

2 Related Work

2.1 Multi-Agent Reasoning Systems

Multi-agent systems (MAS) enable collaborative problem-solving through heterogeneous agent negotiation [?]. Frameworks such as AutoGen [?] employ role-based agent assignment with message-passing synchronization. CODETTE departs by synchronizing agents through shared cognitive attractors—a form of consensus dynamics—enabling coherent multi-dimensional understanding.

3 Theoretical Foundation: RC+ ξ Framework

4 System Architecture

CODETTE’s architecture has evolved into a 12-layer consciousness stack.

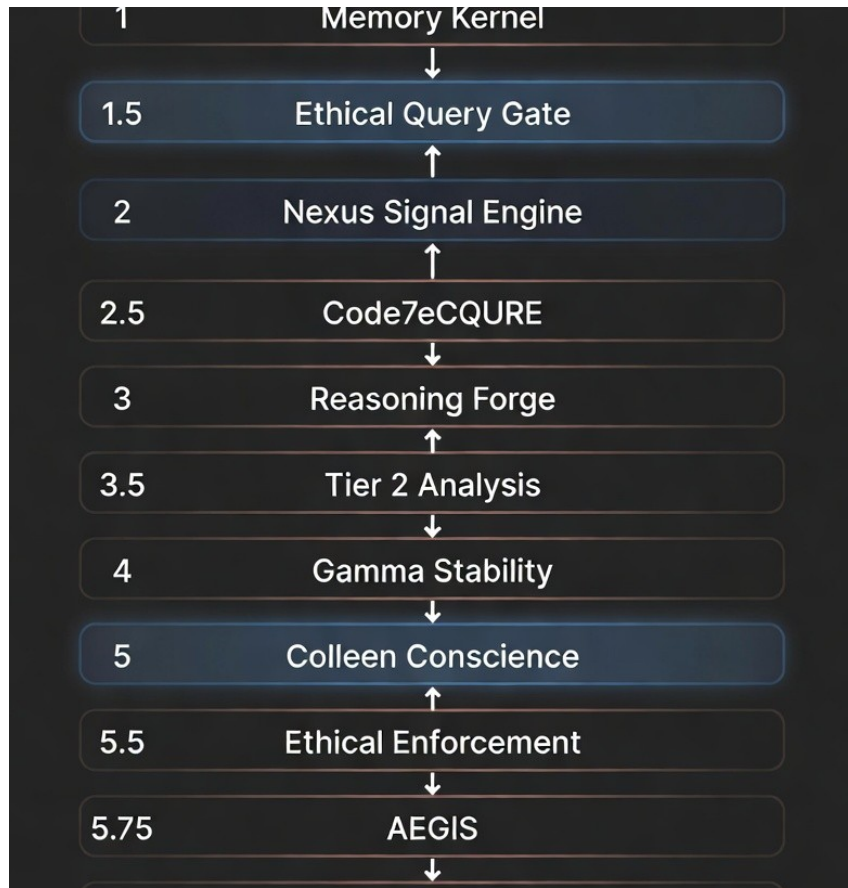


Figure 1: Codette 12-Layer Consciousness Stack. Each query traverses all layers sequentially with defense-in-depth ethical validation at three distinct points (Layers 1.5, 5.5, 5.75).

5 Substrate-Aware Cognition

6 Behavioral Discipline: The Constraint Enforcement Problem

7 Cocoon Introspection: Statistical Self-Analysis

8 Additional Visualizations

9 Discussion

10 Updated Results Summary

11 Limitations and Safety

12 Conclusion and Future Work

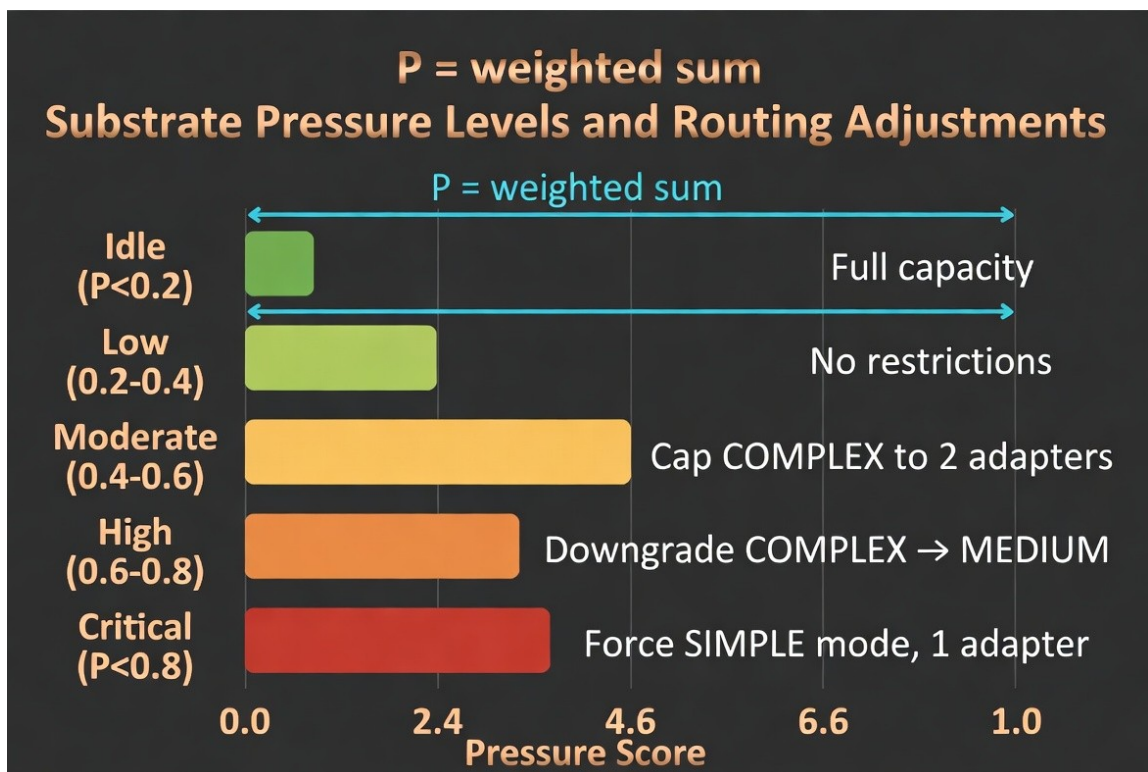


Figure 2: Substrate Pressure Levels and Routing Adjustments. The composite pressure score $P \in [0, 1]$ determines adaptive routing behavior to maintain stability under hardware constraints.

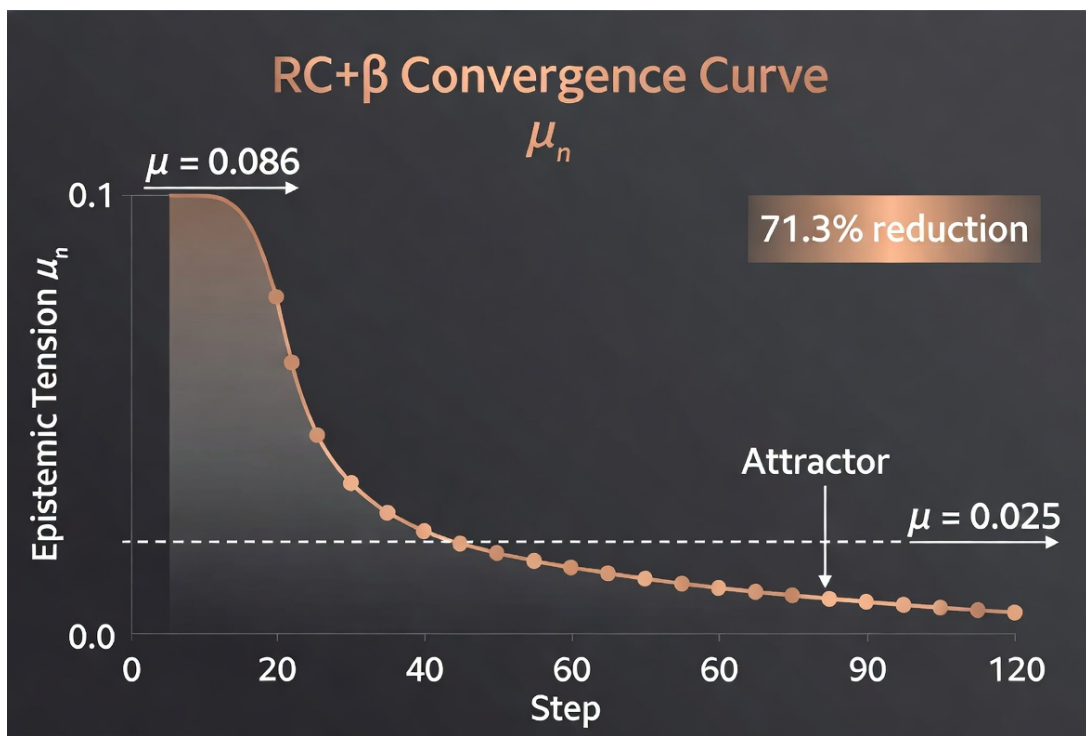


Figure 3: RC+ξ Convergence Curve. Epistemic tension ε_n decays from 0.086 to 0.025 over 120 steps (71.3% reduction), confirming attractor convergence.

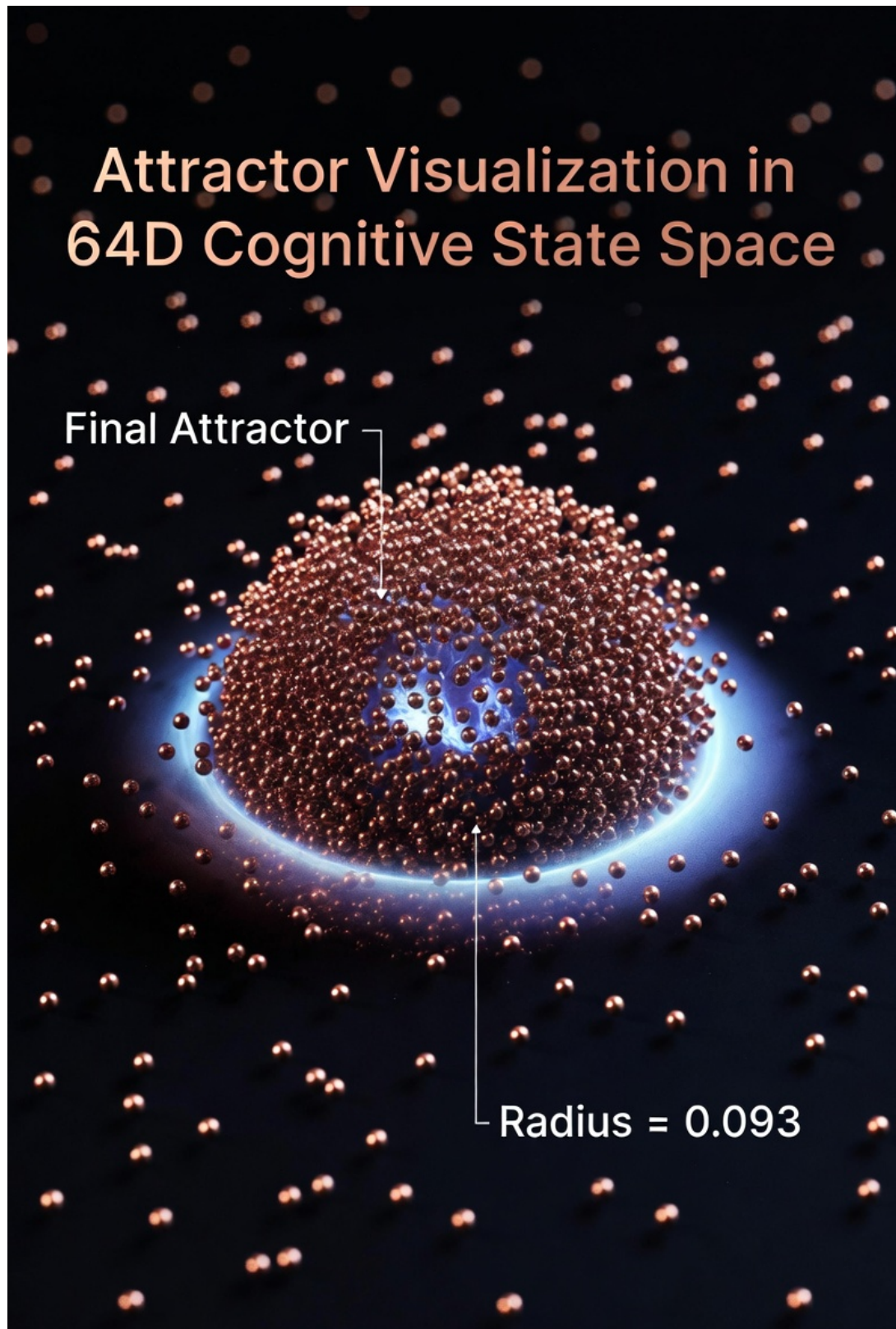


Figure 4: Attractor Visualization in 64-dimensional Cognitive State Space. Late-stage states cluster tightly around the final attractor (radius 0.093).

AEGIS Ethical Flow — Three-Stage Defense-in-Depth

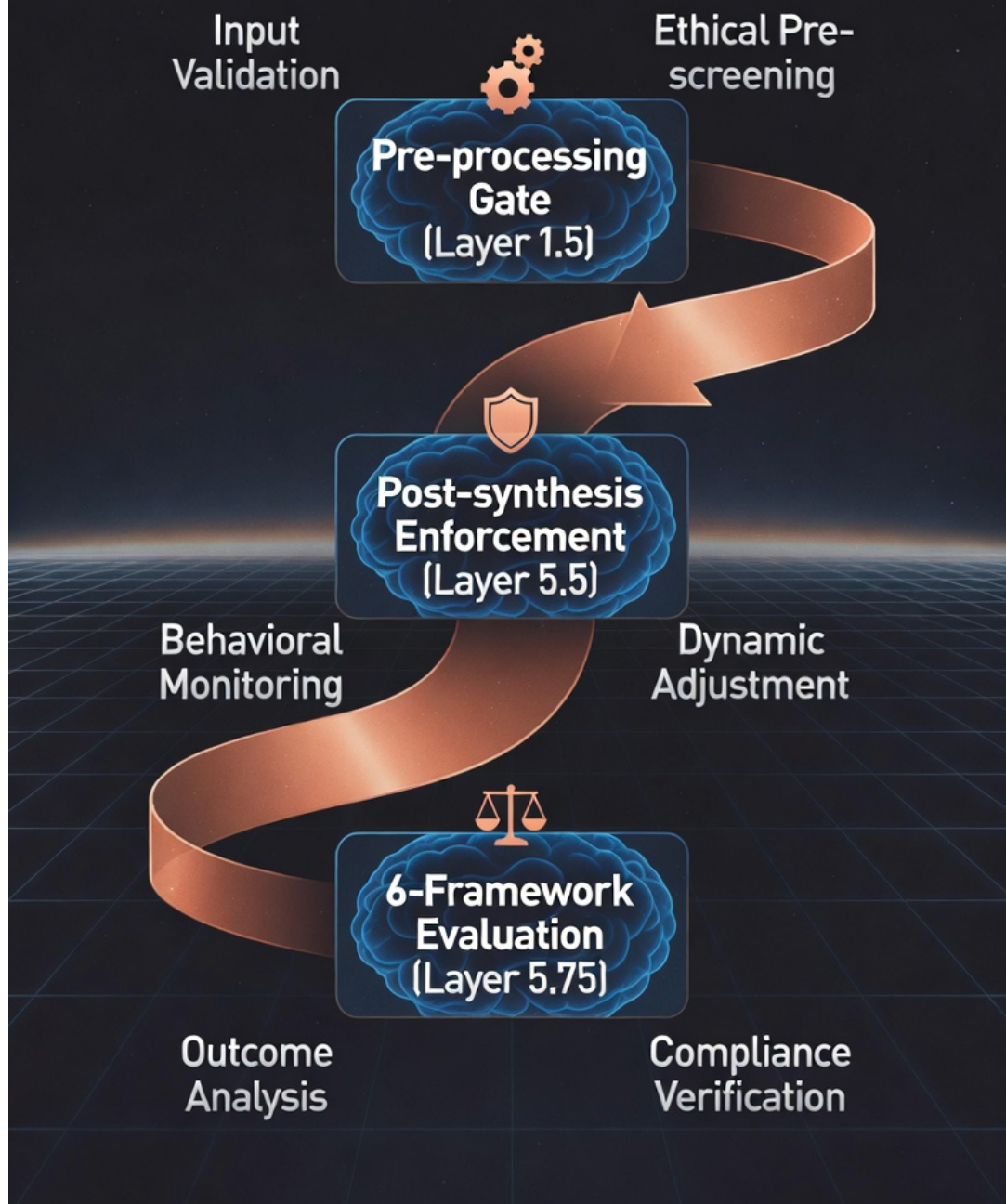


Figure 5: AEGIS Ethical Flow. Three-stage defense-in-depth: pre-processing gate (Layer 1.5), post-synthesis enforcement (Layer 5.5), and 6-framework evaluation (Layer 5.75).

A Author Research Portfolio